

Kolej Matrikulasi Kelantan
DC014 COMPUTER SCIENCE
TRIAL-PSPM 2023/2024
(2 HOURS/70 MARKS)

Name : _____ Class: _____

Matric No : _____

1. Today's world is heavily dependent on multimedia in daily life. All sectors of the profession use multimedia in daily life directly or indirectly to enhance and advance it. As a lecturer of course Ahmad will use graphics as a multimedia element in teaching and learning.

- a. Describe **one (1)** other multimedia element that can be used in a teaching presentation other than graphic.

[2 marks]

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- b. Explain **one (1)** importance of using the element you answer in **1(a)**.

[2 marks]

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- c. Differentiate the **axis** of two-dimensional (2D) and three-dimensional (3D) multimedia representation.

[2 marks]

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- d. Multimedia delivery is important to distribute the content of education. **Two (2)** types of multimedia delivery options are online and offline.

Choose **one (1)** type suitable to deliver content education. Based on the option you choose, justify the advantages of your answer.

[2 marks]

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- e. Explain **one (1)** application of multimedia in the following field:

- i. Business [2 marks]

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- ii. Healthcare [2 marks]

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- f. Navigation is the outline of the connections or links among various areas of your content and helps you organize your content and messages.

- i. Identify the **two(2)** types of navigation usually used in a website

[2 marks]

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- ii. Explain **one(1)** type of the navigation from **1(f)(i)** above.

[2 marks]

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2. Multimedia design is the art of integrating different media, including text, images, audio and video to communicate with an audience.

- a. The elegance and simplicity of the three-phase (3Ps) model have contributed to its widespread adoption as a conceptual framework for the production process.

- i. Describe **one (1)** activity in production phase. [2 marks]

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- ii. Describe **one (1)** activity in post-production phase. [2 marks]

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- b. An authoring tool is a program that enables you to create a digital course and publish it in select formats.

- Describe **one (1)** type of multimedia authoring tools. [2 marks]

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3. Computer technology has been used to facilitate the storage, production and updating of data.

- a. Explain the concept of the database system. [2 marks]

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- b. Describe the advantages of using a database system compared to a file processing system [2 marks]

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- c. Identify the component of database system based on description below
[2 marks]

Description	Component of Database System
i. The instructions and rules that govern the design and use of the database system	
ii. The raw material from which information is generated	

4. Data in a database system has a definite hierarchy and integrity.

- a. Explain the importance of data integrity.

[2 marks]

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- b. Identify the database object based on the description below.

[2 marks]

Description	Database Object
i. A window on the screen that provides areas for entering or modifying data in a database	
ii. A request for specific data from the database	

5. a. Describe the several uses of network below.

- i. Global Positioning System (GPS)

[2 marks]

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- ii. Videoconferencing

[2 marks]

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b. In a network, computers and devices connected together via transmission media and communication devices.

i. Explain transmission media?

[2 marks]

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ii. Identify type of transmission media that carries signal through substances without physical material on a network

[1 mark]

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iii. State **two (2)** examples of transmission media in question **5(b)(ii)** that is appropriate for home network

[2 marks]

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c. i. What is a communication device?

[2 marks]

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- ii. Name and describe the most appropriate **communication device A** in **FIGURE 1**.

[2 marks]

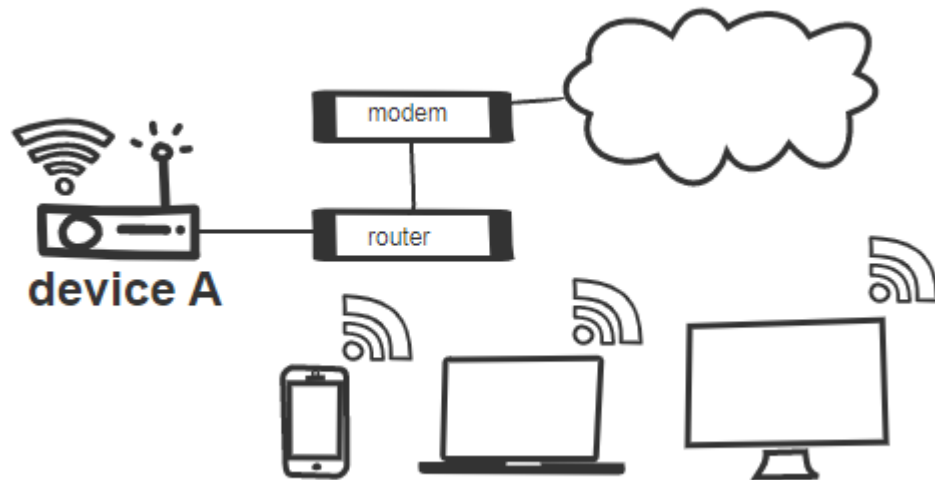


FIGURE 1

- iii. What is the most appropriate communication device to replace communication **device A** if a laptop does not have wireless capability and has to use wired instead?

[1 mark]

- iv. State **two (2)** communication devices other than modem, router and communication **device A** in **FIGURE 1**.

[2 marks]

- v. Describe the function of communication device base on **one of your answer** in **5(c)(iv)**

[2 marks]

6. a. Identify the type of search tool based on the description below.

[2 marks]

Description	Type of search tool
i. The search result is displayed based on hits or popularity	
ii. The search result is displayed based on categories	

- b. Search operators can help to refine search results.

- i. Describe the advantages of using **phrases with search operators** instead of using only phrases while searching information.

[2 marks]

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- ii. State **two (2)** examples of using search operators in searching information.

[2 marks]

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- c. Differentiate between **two (2)** types of web activities and give **one (1)** example of each web activities below.

[6 marks]

	(i) Instant Messaging	(ii) E-commerce
Description		
Example		

d. Information privacy is important to protect us as individuals or businesses.

- i. Describe **two (2)** ways to safeguard our personal information while web browsing.

[2 marks]

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- ii. Explain the importance of safeguarding our personal information while web browsing.

[2 marks]

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e. The Fourth Industrial Revolution (4IR) is a new and emerging technology distinguishing it from previous industrial revolutions.

- i. What is the relationship between 4IR and Internet of Things (IoT).

[2 marks]

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- ii. Discuss **one (1)** example of the Internet of Things (IoT) that is used at home.

[2 marks]

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END OF QUESTION PAPER